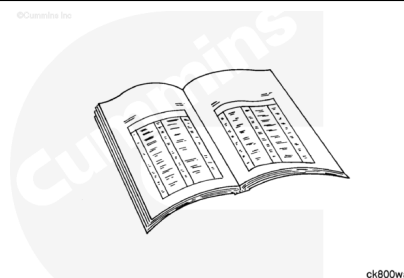


Trainings ▾

Preparatory Steps

- Remove the belt guards, if equipped. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-001.html>)
- Remove the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)
- Remove the fan drive idler arm assembly. Refer to Procedure 008-029 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-029-tr.html>)

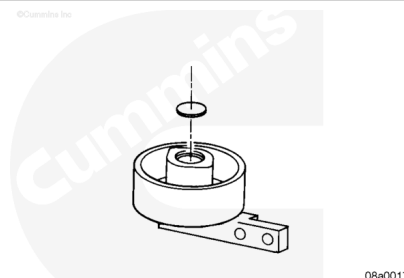


Disassemble

Remove the expansion plug from the pulley.

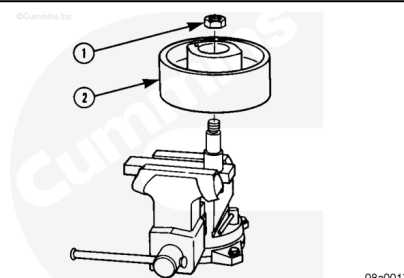
Note : Some pulleys have a plate and gasket instead of an expansion plug.

Remove the plate and gasket.



The locknut has a left hand thread. To loosen the nut turn to the right.

Remove the locknut (1). Lift the pulley (2) from the shaft.

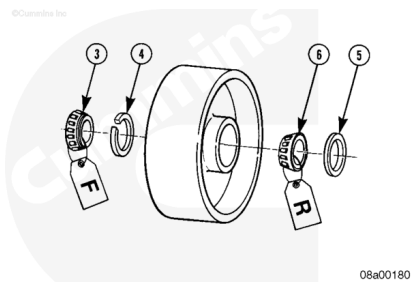


Note : The bearings and spacers are part of a kit. All parts must be replaced, if any part is worn or damaged.

Remove the seal (5).

Remove the bearings (3) and (6). Mark them for location.

Remove the inner bearing spacer (4).

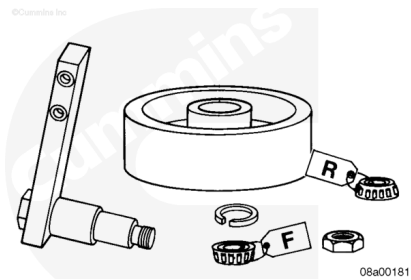


Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not allow the force of the compressed air to turn the bearing when cleaning. Bearing damage will result due to lack of lubrication.

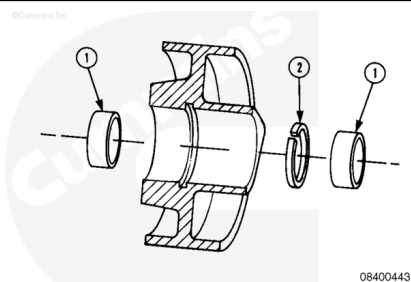
Clean and check the parts.



Use a hammer and a drift to remove the bearing races (1).

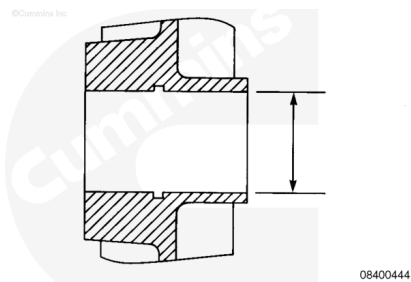
Remove the retaining ring (2).

The thickness of the retaining ring has an affect on the bearing end clearance. The bearing kit contains a new retaining ring.



Measure the inside diameter.

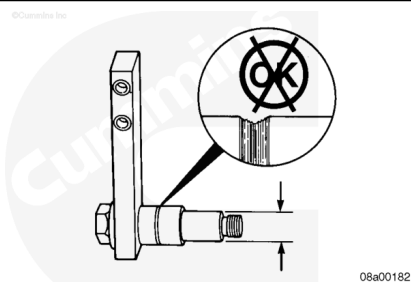
Pulley Bore Inside Diameter		
mm		in
45.936	MIN	1.8085
45.961	MAX	1.8095



Check the shaft for wear at the seal location.

Measure the outside diameter.

Shaft Outside Diameter		
mm		in
21.961	MIN	0.8646



Shaft Outside Diameter		
mm		in
21.974	MAX	0.8651

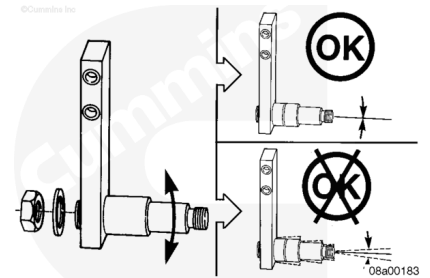
If the shaft shows any wear at the seal surface, the shaft assembly **must** be replaced.

Remove the shaft locknut and the lockwasher.

Be sure the shaft is tight in the bracket. If the shaft is loose, replace both pieces.

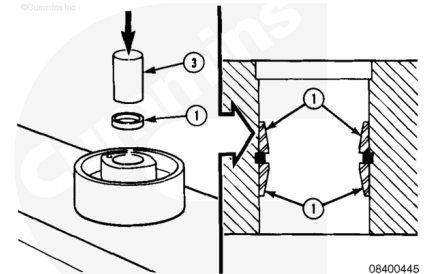
Install the parts.

Torque Value: 150 n·m [110 ft-lb]



Install the retaining ring.

Use a mandrel (3) and an arbor press to install the bearing races (1). They **must** be in tight contact with the retaining ring.



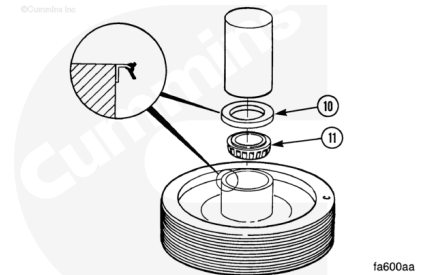
⚠ CAUTION ⚠

Do not lubricate the seal or the seal surface of the shaft. The seal and seal surface must be clean and dry.

Use water pump type grease to lubricate the rear bearings (11) and the bearing race.

Place the bearing on the race.

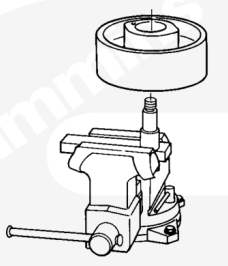
Use a mandrel (7) to install the seal (10). The seal **must not** be more than 0.51 mm [0.020 in] below the top surface.



Assemble

Install the pulley on the shaft.

The bearing **must** slide easily over the shaft. If it does **not**, check for chips or burrs.



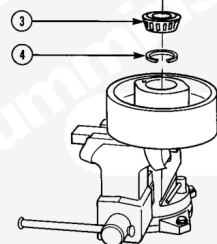
08a00185

Install the inner bearing spacer (4).

Use water pump type grease to lubricate the bearing and the bearing race.

Install the bearing (3).

©Cummins Inc.



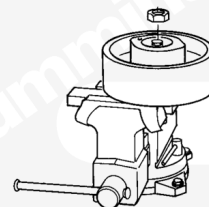
08a00186

The locknut has a left hand thread. To tighten the locknut, turn to the left.

Install the locknut.

Torque Value: 150 n•m [110 ft-lb]

©Cummins Inc.

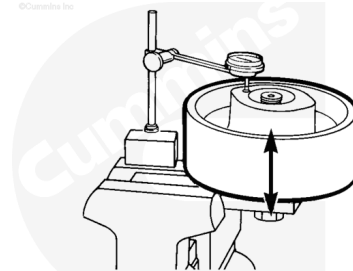


08a00187

Use a dial indicator to measure the clearance.

©Cummins Inc.

Bearing End Clearance		
mm		in
0.08	MIN	0.003
0.25	MAX	0.010



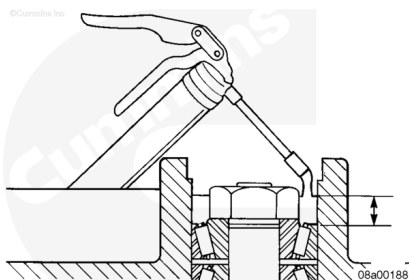
08a00184

If the clearance is **not** within specifications, be sure the spacer has been installed correctly.

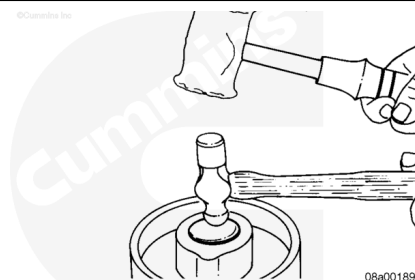
Remove the locknut and inspect the assembly.

The bearings **must** touch the spacer. The bearing races **must** touch the retaining ring.

Fill the front bearing cavity 2/3-full with a water pump type grease.

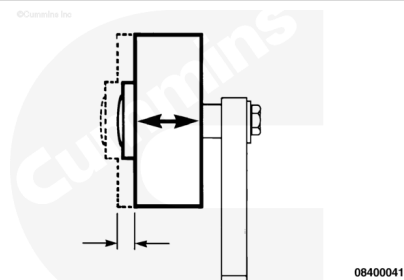


Install the sealing plug or plate.



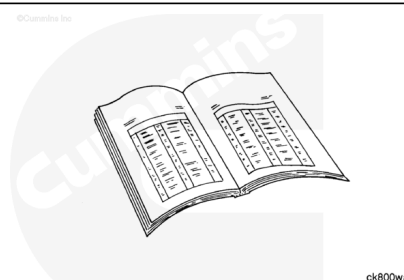
Inspect

Every 6000 hours or 2 years, inspect the fan drive idler pulley assembly. Rebuild or replace the idler pulley as necessary. Contact a Cummins® Authorized Repair Location.



Finishing Steps

- Install the fan drive idler arm assembly. Refer to Procedure 008-029 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-029-tr.html>)
- Install the fan belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-002-tr.html>)
- Install the fan guards, if equipped. Refer to Procedure 008-001 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-001-tr.html>)



- Operate the engine and check for noise.

Last Modified: 05-Nov-2013
